

Qubit-centric Transformer for Surface Code Decoding

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For reliable large-scale quantum computation, quantum error correction (QEC) is essential to protect logical information distributed across multiple physical qubits. Taking advantage of recent advances in deep learning, neural network-based decoders have emerged as a promising approach to improve the reliability of QEC. We propose the qubit-centric transformer (QCT), a novel and universal QEC decoder based on a transformer architecture with a qubit-centric attention mechanism. Our decoder transforms input syndromes from the stabilizer domain into qubit-centric tokens via a specialized embedding strategy. These qubit-centric tokens are processed through attention layers to effectively identify the underlying logical error. Furthermore, we introduce a graph-based masking method that incorporates the topological structure of quantum codes, enforcing attention toward relevant qubit interactions. Across various code distances for surface codes, QCT achieves state-of-the-art decoding performance, significantly outperforming existing neural decoders and the belief propagation (BP) with ordered statistics decoding (OSD) baseline. Notably, QCT achieves a high threshold of 18.1% under depolarizing noise, which closely approaches the theoretical bound of 18.9% and surpasses both the BP+OSD and the minimum-weight perfect matching (MWPM) thresholds. This qubit-centric approach provides a scalable and robust framework for surface code decoding, advancing the path toward fault-tolerant quantum computing.

I. INTRODUCTION

Quantum error correction (QEC) is fundamental to the realization of fault-tolerant quantum computing, as it preserves the integrity of logical information against environmental noise and operational imperfections [1–7]. The QEC process consists of two main stages: encoding, which maps k logical qubits into $n > k$ physical qubits, and decoding, which estimates errors from measured syndromes. Consequently, the performance of the QEC decoder is a primary factor in determining the overall reliability of quantum computation.

Although classical error correction techniques are well-developed [8–10], they cannot be applied directly to QEC. Unlike classical bits, quantum states cannot be copied to generate redundancy due to the no-cloning theorem [2]. Moreover, direct measurement of qubits collapses their quantum state, and qubits are also susceptible not only to bit-flip errors but also phase-flip errors [2]. These challenges necessitate a fundamentally different approach based on indirect stabilizer measurements. This process produces syndromes, from which the underlying qubit errors are inferred without disturbing the encoded quantum information [5].

Recent advances in neural networks have opened a new avenue for QEC decoding [11–17]. Neural network-based decoders are generally categorized into two types: low-level decoders, which reconstruct the physical error pattern of length n , and high-level decoders, which estimate

the logical error pattern of length k . For high-level decoders, the task is formulated as a classification problem over 4^k possible logical error classes. This approach is particularly advantageous for codes with a small k , such as the surface code ($k = 1$). Since the output dimension remains fixed at 4^k regardless of the code distance d , these decoders provide excellent structural scalability without an exponential increase in output complexity [18].

Furthermore, since neural network-based decoders process syndromes through a fixed model architecture, they offer a deterministic decoding latency. This deterministic runtime contrasts with hybrid decoding architectures that employ a pre-decoder and a post-decoder, such as the belief propagation (BP) with ordered statistics decoding (OSD). [19–21]. In these hybrid approaches, the overall decoding time fluctuates depending on the input syndrome pattern, as the computationally intensive post-processing stage is triggered only when the initial pre-decoder fails to converge. Predictable latency is a critical prerequisite for fault-tolerant quantum computing, as the global system clock is limited by the decoder’s worst-case response time [13]. Given the remarkable success of neural networks in classification tasks [22, 23], they have emerged as a promising solution for high-level QEC decoders. Early implementations utilizing feed-forward neural networks (FFNNs) [13] have since evolved into more sophisticated architectures, such as convolutional neural networks (CNNs) [18], to better exploit the spatial structure of quantum codes.

More recently, the transformer architecture has emerged as the universal backbone of modern deep learn-

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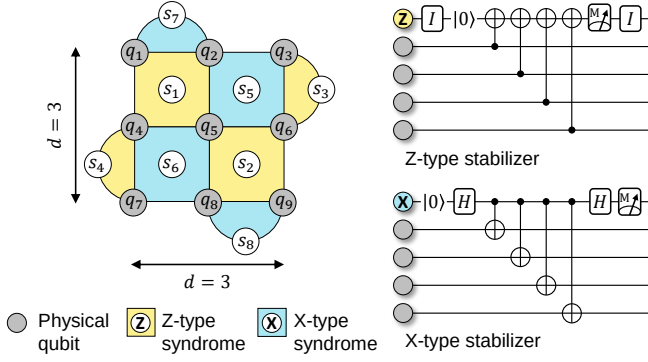


FIG. 1: Illustration of the topological structure of the surface code with distance $d = 3$ (left) and the syndrome extraction circuits (right).

ing, demonstrating state-of-the-art performance across virtually all domains. Leveraging this widespread success, transformer-based decoders [24, 25] have recently been adapted for QEC. For instance, Transformer-QEC [17] utilizes a vision transformer architecture to predict local errors, serving as a pre-decoding stage that reduces syndromes for a global decoder like minimum weight perfect matching (MWPM) [26]. In addition, AlphaQubit utilizes a hybrid architecture combining CNNs and RNNs within a transformer framework, effectively capturing both the topological structure of surface codes and the temporal correlations inherent in repeated syndrome measurements [16].

However, a common limitation of these approaches is their stabilizer-centric input representation, which primarily operates on the syndrome space. While this approach aligns with the syndrome extraction process, it is fundamentally limited as it does not directly represent the physical qubits where the underlying physical errors lead to logical errors. Since logical errors fundamentally arise from physical faults rather than the syndromes themselves, a stabilizer-centric view provides only an indirect and often insufficient perspective. Consequently, a qubit-centric perspective is required to provide a more direct and effective means of analyzing the underlying error patterns.

In this work, we propose the *qubit-centric transformer* (QCT), an architecture that shifts the decoding perspective from stabilizers to physical qubits. Unlike conventional stabilizer-centric approaches, QCT structures the input through two key layers: a qubit embedding layer and a merging layer. The embedding layer projects localized syndrome vectors, which encapsulate the measurement outcomes of adjacent Z- and X-stabilizers, into dense feature tokens. Subsequently, the merging layer concatenates these decoupled representations and fuses them into unified qubit tokens. This design allows the transformer blocks to process tokens that directly correspond to the physical qubits where errors actually originate, demonstrating that a physical qubit-level perspec-

tive is fundamentally effective for decoding. Furthermore, we introduce a *structure-aware mask matrix* that reflects the topological structure of QEC codes. To ensure the model captures the physical structure, this matrix restricts the self-attention mechanism by masking the connections between tokens if their corresponding physical qubits do not share at least one common stabilizer. By incorporating this local structure, QCT focuses on physically relevant neighboring qubits and efficiently learns complex, localized error correlations.

We evaluate the decoding performance of the proposed QCT on surface codes across various code distances. Consequently, we demonstrate that QCT achieves substantially lower logical error rates (LER) and higher pseudo-thresholds compared to existing neural network decoders. Notably, our analysis reveals that QCT achieves a high threshold of approximately 18.1% under depolarizing noise, which closely approaches the theoretical bound of 18.9% [27]. This performance significantly outperforms the 17.00% threshold of the BP+OSD baseline and the 14.7% threshold of the conventional MWPM algorithm. These results validate the robustness of the QCT and its potential for large-scale fault-tolerant quantum computing.

The rest of the paper is structured as follows. In Sec. II, we briefly review the fundamentals of stabilizer codes, surface codes, and the high-level decoding method. Then, in Sec. III, we present the proposed QCT architecture, including qubit-centric embedding, merging layer, and structure-aware masking strategy. In Sec. IV, we evaluate the performance of our model, comparing its logical error rate against several classical and neural network-based decoders. Furthermore, we provide an analysis of the architectural features and evaluate both pseudo-thresholds and the threshold. Finally, Sec. V concludes the paper.

II. PRELIMINARIES

A. Stabilizer code

QEC codes utilize n physical qubits to encode k logical qubits. An $[[n, k, d]]$ stabilizer code $\mathcal{C}$ is defined by a stabilizer group $\mathcal{S}$, which is an Abelian subgroup of the n -qubit Pauli group $\mathcal{P}_n$ such that $-I_n \notin \mathcal{S}$ [5]. The Pauli group $\mathcal{P}_n$ consists of n -fold tensor products of the single-qubit Pauli operators $\mathcal{P} = \{I, X, Z, Y\}$, defined as:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, Y = iXZ.$$

The group $\mathcal{S}$ is generated by $n - k$ independent and commuting stabilizer generators $\{S_1, S_2, \dots, S_{n-k}\}$. The code space $\mathcal{C}$ is then defined as the common +1 eigenspace of all elements in $\mathcal{S}$:

$$\mathcal{C} = \{|\psi\rangle \mid S|\psi\rangle = |\psi\rangle, \forall S \in \mathcal{S}\}.$$

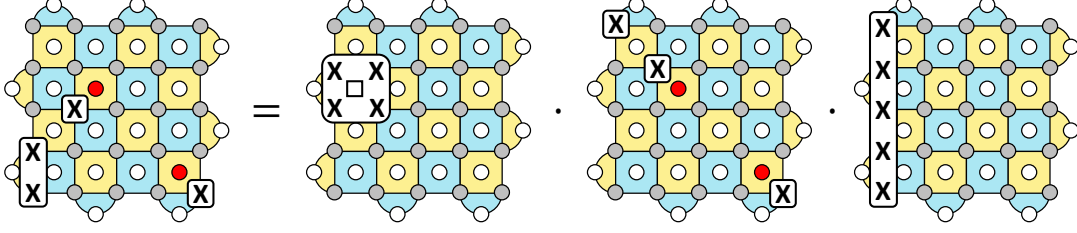


FIG. 2: Illustration of the high-level decoding methodology treating decoding as a classification problem. An arbitrary physical error E can be decomposed into $E = S \cdot T \cdot L$.

Any state $|\psi\rangle \in \mathcal{C}$ is referred to as a logical state, representing the collective state of the k encoded logical qubits. By definition, stabilizers $S \in \mathcal{S}$ act trivially on all logical states, leaving the encoded information unchanged.

Logical operators are Pauli operators that commute with all elements in $\mathcal{S}$, thereby preserving the code space $\mathcal{C}$. Specifically, non-trivial logical operators are those contained in the centralizer of $\mathcal{S}$ but not in $\mathcal{S}$ itself, which can be defined as:

$$\mathcal{L} = \{L \mid [L, S] = 0, \forall S \in \mathcal{S}\} / \mathcal{S}, \quad (1)$$

where $[L, S] = LS - SL$ denotes the commutator of operators L and S . To distinguish them from physical operators, these operators acting on the encoded logical space are conventionally denoted as $L \in \{\bar{I}, \bar{X}, \bar{Z}, \bar{Y}\}$. Here, the logical identity $\bar{I}$ acts trivially on the logical state (equivalent to the elements of $\mathcal{S}$), while the non-trivial logical operators $\bar{X}, \bar{Z}$, and $\bar{Y}$ act non-trivially on the logical qubits by transforming one logical state into another. Since any stabilizer $S \in \mathcal{S}$ acts trivially on the code space, multiplying a logical operator L by a stabilizer S yields an equivalent logical operator (i.e., $(LS)|\psi\rangle = L(S|\psi\rangle) = L|\psi\rangle$). The code distance d is defined as the minimum weight of any such non-trivial logical operator.

When an error $E \in \mathcal{P}_n$ occurs on the physical qubits, it is identified by measuring the eigenvalues of the stabilizer generators. This process yields a syndrome vector $\mathbf{s} = (s_1, \dots, s_{n-k}) \in \{0, 1\}^{n-k}$, where each syndrome bit s_i is defined as:

$$s_i = \begin{cases} 0 & \text{if } [S_i, E] = 0, \\ 1 & \text{if } [S_i, E] \neq 0. \end{cases} \quad (2)$$

B. Surface Codes

The surface code is a class of topological stabilizer codes, widely recognized for its high error threshold and its leveraging of local connectivity between qubits [28, 29]. As illustrated in Fig. 1, which shows a rotated

surface code with distance $d = 3$, physical qubits (often referred to as data qubits, indicated by grey circles) are arranged on the vertices of a $d \times d$ lattice to encode $k = 1$ logical qubit ($n = d^2$). To detect physical errors, $m = n - k = d^2 - 1$ stabilizer generators are distributed across the faces (plaquettes) of the lattice. For each face, an ancilla qubit is positioned to measure a syndrome bit. The right panel of Fig. 1 illustrates the syndrome extraction circuit for this measurement. These stabilizers are categorized into Z-stabilizers (yellow plaquettes) and X-stabilizers (blue plaquettes), which are used to detect X (bit-flip) and Z (phase-flip) errors on the physical qubits, respectively.

The measurement outcomes of these stabilizers form the full syndrome vector $\mathbf{s} = (s_1, \dots, s_m) \in \{0, 1\}^m$. Following the indexing order shown in Fig. 1—where degree-4 stabilizers are indexed prior to degree-2 stabilizers—the vector $\mathbf{s}$ is partitioned as $\mathbf{s} = [\mathbf{s}^Z, \mathbf{s}^X]$. Specifically, the first m^Z bits $\mathbf{s}^Z = (s_1, \dots, s_{m^Z})$ correspond to the Z-stabilizers, and the subsequent m^X bits $\mathbf{s}^X = (s_{m^Z+1}, \dots, s_m)$ correspond to the X-stabilizers ($m = m^Z + m^X$).

On this topological structure, logical operators are defined as non-trivial Pauli operators that span across the lattice. Specifically, the logical $\bar{Z}$ operator corresponds to a chain of physical Z operators connecting the top and bottom boundaries, while the logical $\bar{X}$ operator corresponds to a chain of physical X operators connecting the left and right boundaries. Together with the logical identity $\bar{I}$ and $\bar{Y} \equiv i\bar{X}\bar{Z}$, these constitute the fundamental logical operators of the surface code.

C. High-level decoding method

We adopt a high-level decoding method [13, 18] based on a neural decoding architecture, which simplifies the learning task as shown in Fig. 2. This approach exploits the decomposition of any physical error E into three distinct components: $E = S \cdot T \cdot L$. Here, S is a stabilizer, L is a logical operator, and T is a pure error that can be deterministically inferred from the syndrome $\mathbf{s}$. For a fixed E and T , the stabilizer S and the logical operator L

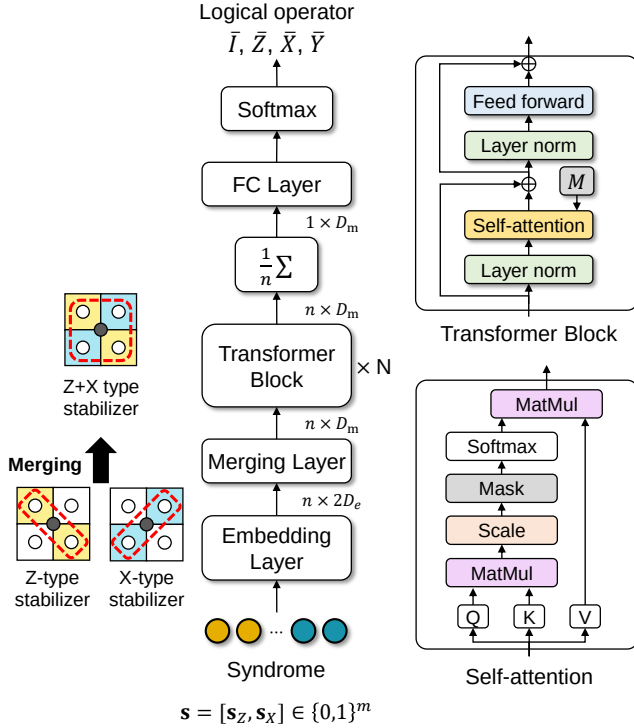


FIG. 3: The proposed QCT architecture. QCT processes the input syndrome vector $\mathbf{s}$ through the embedding layer and merging layer, followed by N transformer blocks, to finally predict the logical operator.

are uniquely determined, implying that each error E can be mapped to a corresponding logical operator L . For surface codes, since it encodes exactly one logical qubit ($k = 1$), the logical operator is strictly restricted to the four equivalence classes, i.e., $L \in \{\bar{I}, \bar{X}, \bar{Z}, \bar{Y}\}$. Thus, the objective of the neural decoder f_θ , where θ denotes the model parameters, is reduced to a four-class classification problem: predicting the most likely logical operator $\hat{L} = f_\theta(\mathbf{s})$ from the syndrome vector $\mathbf{s}$. The final recovery operator is determined as $R = T \cdot \hat{L}$. If the prediction is correct ($\hat{L} = L$), the combined operation of error and recovery results in $RE = (TL)(STL) = S$, which preserves the code state since $RE|\psi\rangle = S|\psi\rangle = |\psi\rangle$ for any $|\psi\rangle \in \mathcal{C}$.

In contrast to low-level decoding, which aims to identify the exact physical error E among 4^n possible outcomes, this high-level decoding approach fixes the output dimension as k regardless of the code size n . Therefore, the model scales for length n efficiently without a rapid increase in structural complexity or training difficulty.

III. QUBIT-CENTRIC TRANSFORMER

In this section, we introduce QCT architecture, which consists of two key components: a *qubit-centric embed-*

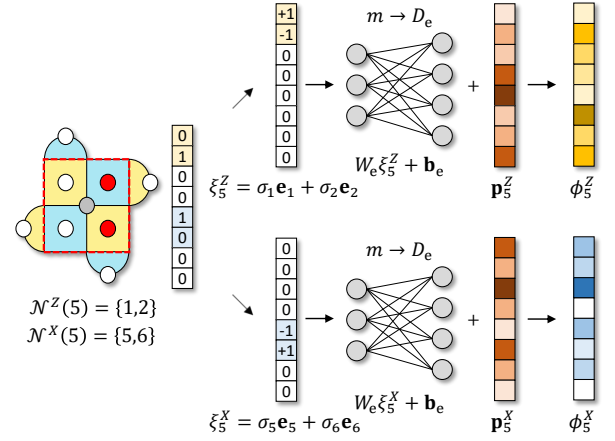


FIG. 4: Illustration of the embedding process for physical qubit q_5 for $d = 3$ surface code.

ding layer and a merging layer. These layers extract decoupled Z- and X-syndrome information for each physical qubit and subsequently fuse them into unified sequence tokens. These tokens are then processed through attention layers to learn the complex error correlations. Furthermore, we incorporate a *structure-aware mask matrix* that constrains the self-attention mechanism to reflect the topological structure of the QEC code. The overall architecture of the proposed QCT is illustrated in Fig. 3.

A. Qubit-centric Embedding Layer

Let $\mathcal{N}^Z(i)$ and $\mathcal{N}^X(i)$ denote the sets of indices for Z- and X-stabilizers adjacent to the physical qubit q_i , respectively. For example, as illustrated in Fig. 1, a central physical qubit q_5 is adjacent to two Z-stabilizers, $\mathcal{N}^Z(5) = \{1, 2\}$, and two X-stabilizers, $\mathcal{N}^X(5) = \{5, 6\}$. Due to the topological structure of the surface code, each physical qubit connects to at most two stabilizers of each type, i.e., $|\mathcal{N}^Z(i)| \leq 2$ and $|\mathcal{N}^X(i)| \leq 2$.

Let $\{\mathbf{e}_j\}_{j=1}^m$ be the standard basis vectors of $\mathbb{R}^m$ (e.g., $\mathbf{e}_1 = [1, 0, \dots, 0]^\top$) and let $\sigma_j = (-1)^{s_j} \in \{+1, -1\}$ denote the value mapped from the j -th syndrome bit s_j . Throughout this paper, all vectors are defined as column vectors unless otherwise specified. For a stabilizer type $\tau \in \{Z, X\}$, the localized syndrome vector ξ_i^τ is defined as a sparse representation of the syndrome vector $\mathbf{s}$, localized to the neighbors of qubit q_i :

$$\xi_i^\tau = \sum_{j \in \mathcal{N}^\tau(i)} \sigma_j \mathbf{e}_j. \quad (3)$$

For each physical qubit q_i , we construct two localized syndrome vectors ξ_i^Z and ξ_i^X of size m based on the equation (3), which encapsulate the measurement outcomes of the adjacent stabilizers. Note that the elements corresponding to non-adjacent stabilizers (i.e., $j \notin \mathcal{N}^\tau(i)$) are zero, ensuring that each qubit-centric vector carries

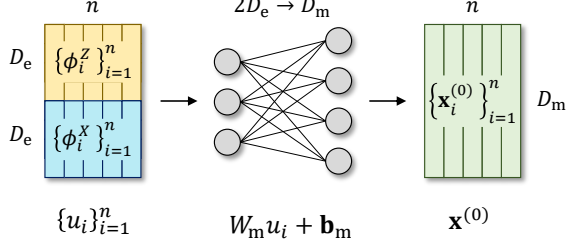


FIG. 5: Illustration of the merging Layer. The input consists of n decoupled Z-tokens $\{\phi_i^Z\}_{i=1}^n$ (yellow) and n decoupled X-tokens $\{\phi_i^X\}_{i=1}^n$ (blue) separately. This process fuses the decoupled embeddings into unified qubit tokens $\mathbf{x}^{(0)}$.

only local syndrome information relevant to the specific physical qubit.

Fig. 4 illustrates this embedding process for the physical qubit q_5 in a $d = 3$ surface code. In this example, we assume that the syndrome bits s_2 and s_5 are 1, while all others are 0. Since q_5 is adjacent to Z-stabilizers indexed as $\mathcal{N}^Z(5) = \{1, 2\}$ and X-stabilizers as $\mathcal{N}^X(5) = \{5, 6\}$, localized syndrome vectors for q_5 are constructed as:

$$\xi_5^Z = \sigma_1 \mathbf{e}_1 + \sigma_2 \mathbf{e}_2, \quad \xi_5^X = \sigma_5 \mathbf{e}_5 + \sigma_6 \mathbf{e}_6. \quad (4)$$

Given the syndrome $s_2 = 1$ and $s_5 = 1$, we have $\sigma_2 = -1$ and $\sigma_5 = -1$, while the remaining coefficients are $\sigma_1 = +1$ and $\sigma_6 = +1$. Note that ξ_5^Z is a sparse vector since most of its elements are zero.

Subsequently, each sparse localized syndrome vector is projected into a dense feature to extract high-level representations. We first project ξ_i^Z and ξ_i^X into D_e -dimensional vectors via a fully connected (FC) layer and add learnable positional embedding vectors $\mathbf{p}_i^Z, \mathbf{p}_i^X \in \mathbb{R}^{D_e}$, which are optimized during training. Let ϕ_i^Z and ϕ_i^X denote the resulting feature embeddings for the Z- and X-type syndromes of the i -th qubit, respectively:

$$\phi_i^\tau = (W_e \xi_i^\tau + \mathbf{b}_e) + \mathbf{p}_i^\tau, \quad (5)$$

where $W_e \in \mathbb{R}^{D_e \times m}$ and $\mathbf{b}_e \in \mathbb{R}^{D_e}$ are the weights and biases of the FC layer. Finally, as shown in Fig. 4, the feature embeddings ϕ_5^Z and ϕ_5^X for the example qubit are obtained by applying this transformation to the constructed vectors ξ_5^Z and ξ_5^X .

B. Merging Layer

Finally, a *merging layer* integrates these decoupled representations of $\{\phi_i^Z\}_{i=1}^n$ and $\{\phi_i^X\}_{i=1}^n$ into a unified form. For each qubit q_i , we first concatenate the corresponding Z- and X-type embeddings into paired vectors $\{u_i\}_{i=1}^n$, which are defined as:

$$u_i = \begin{bmatrix} \phi_i^Z \\ \phi_i^X \end{bmatrix} \in \mathbb{R}^{2D_e}, \quad (6)$$

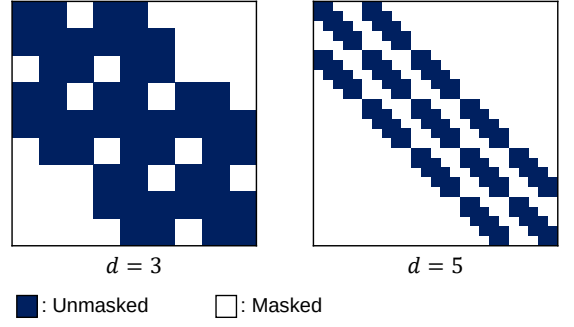


FIG. 6: Illustration of the structure-aware masks M for surface codes used in QCT. The left figure shows the mask for $d = 3$ surface code, and the right figure shows the mask for $d = 5$ surface code.

where the notation above denotes the vertical concatenation (stacking) of the two column vectors along the feature dimension.

Then, we fuse the decoupled representations by projecting u_i into a unified token $\mathbf{x}_i^{(0)}$ of dimension D_m via a simple merging layer:

$$\mathbf{x}_i^{(0)} = W_m u_i + \mathbf{b}_m, \quad (7)$$

where $W_m \in \mathbb{R}^{D_m \times 2D_e}$ and $\mathbf{b}_m \in \mathbb{R}^{D_m}$ are trainable parameters. The resulting qubit-centric tokens $\{\mathbf{x}_i^{(0)}\}_{i=1}^n$ serve as the initial input to the transformer. Fig. 5 illustrates the merging process of QCT.

C. Transformer and Output Layer

These tokens are concatenated to form the input matrix $\mathbf{x}^{(0)} = [\mathbf{x}_1^{(0)}, \dots, \mathbf{x}_n^{(0)}] \in \mathbb{R}^{D_m \times n}$. As shown in Fig. 3, each layer $l \in \{1, \dots, N\}$, the transformer block generates the output sequence $\mathbf{x}^{(l)}$ and uses them as the next input sequence. Each block consists of a self-attention module, a feed-forward neural network (FFNN), and normalization layers.

The self-attention mechanism is parameterized by three learnable projection matrices for the query, key, and value: $W_q \in \mathbb{R}^{D_m \times D_h}$, $W_k \in \mathbb{R}^{D_m \times D_h}$, and $W_v \in \mathbb{R}^{D_m \times D_h}$, respectively, where D_h is the dimension of each head. Given an input X , the projections are computed as $Q = XW_q$, $K = XW_k$, and $V = XW_v$. Finally, the self-attention output is obtained by the scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{D_h}} \right) V. \quad (8)$$

The transformer blocks capture the global dependencies and relationships between qubits via the multi-head self-attention mechanism.

Finally, the output tokens from this block are aggregated into a single vector via mean pooling and passed

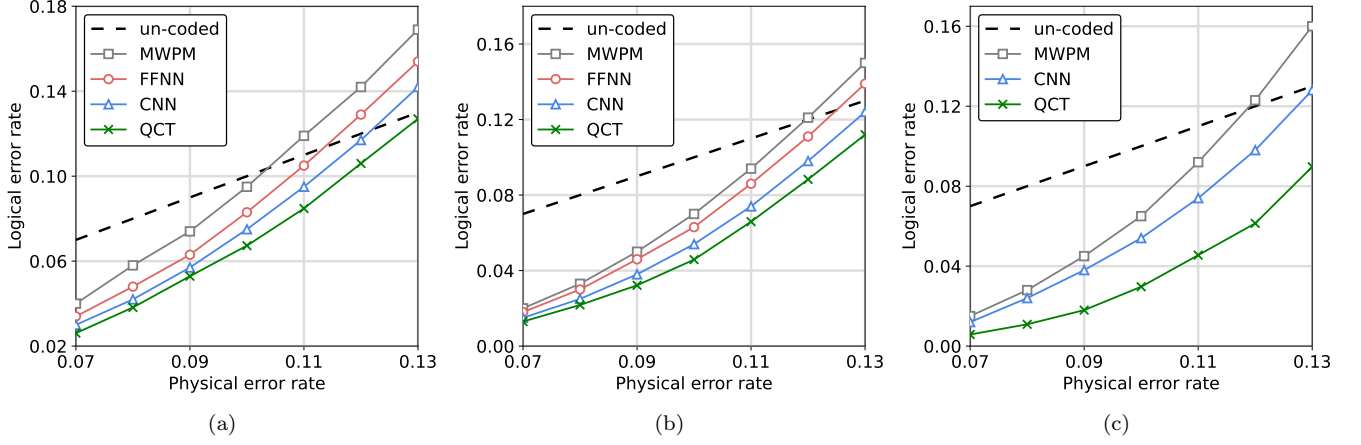


FIG. 7: Logical error rate as a function of the physical error rate for the surface code under a depolarizing noise model. The performance of the proposed QCT is benchmarked against MWPM, FFNN, and CNN decoders for code distances (a) $d = 5$, (b) $d = 7$, and (c) $d = 9$.

through a FC layer to obtain the final prediction. For surface codes, since $k = 1$, QCT outputs a 4-dimensional logit vector. Applying a softmax function to these logits produces a probability distribution $\hat{P}_c$ for class c over the four logical error classes $c \in \{\bar{I}, \bar{X}, \bar{Z}, \bar{Y}\}$. The class with the highest probability is selected as the decoder's prediction.

The entire QCT model is trained end-to-end by minimizing the standard cross-entropy loss, $\mathcal{L}_{CE}$, between the predicted and true logical operators. The loss is defined as:

$$\mathcal{L}_{CE} = - \sum_{c \in \{\bar{I}, \bar{X}, \bar{Z}, \bar{Y}\}} y_c \log(\hat{P}_c), \quad (9)$$

where y_c is the one-hot encoded true label (1 if c is the correct class, 0 otherwise).

D. Structure-aware Mask Matrix

To learn the topological structure of the surface codes and improve learning efficiency, we propose a structure-aware mask for QCT. In our architecture, each input token represents a unified physical qubit q_i , encapsulating its syndrome information. Reflecting the structure of the surface code, the structure-aware mask M is designed to allow any two qubits q_i and q_j to attend to each other if and only if they share at least one common stabilizer, regardless of whether it is a Z- or X-stabilizer. All other qubit pairs that share no stabilizers are masked, enabling the model to effectively capture the local structure of the codes.

Let $\mathcal{N}(i)$ be the set of all stabilizers adjacent to qubit q_i , including all Z- and X-stabilizers adjacent to it:

$$\mathcal{N}(i) = \mathcal{N}^Z(i) \cup \mathcal{N}^X(i), \quad (10)$$

where $\mathcal{N}^Z(i)$ and $\mathcal{N}^X(i)$ denote the sets of adjacent Z- and X-stabilizers for q_i , respectively. We then define $\mathcal{R}(i)$ as the index set of all qubits q_j (including q_i itself) that share at least one stabilizer with q_i :

$$\mathcal{R}(i) = \{j \mid \mathcal{N}(j) \cap \mathcal{N}(i) \neq \emptyset\}. \quad (11)$$

Each element of the mask matrix $M \in \{0, -\infty\}^{n \times n}$ is then defined as:

$$M_{ij} = \begin{cases} 0 & \text{if } j \in \mathcal{R}(i), \\ -\infty & \text{otherwise.} \end{cases} \quad (12)$$

Applying this mask modifies the self-attention calculation as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{D_h}} + M\right) V. \quad (13)$$

Fig. 6 shows the structure-aware mask matrix for $d = 3$ and $d = 5$ surface codes. White entries represent masked relations ($-\infty$) and blue entries represent unmasked relations (0). This mask enforces the local connectivity of the surface code by allowing tokens to attend to each other only if their corresponding physical qubits share at least one common stabilizer (either Z-type or X-type).

IV. PERFORMANCE EVALUATION

A. Comparison of Decoding Performance

We evaluate the LER performance of the proposed QCT for the surface codes under the depolarizing noise model. In all experiments, we choose transformer blocks with $N = 8$ layers, $D_e = 64$, and $D_m = 128$. We first benchmark our model against the classical MWPM algorithm [30] and two representative neural network architectures: the FFNN [13] and CNN [18].

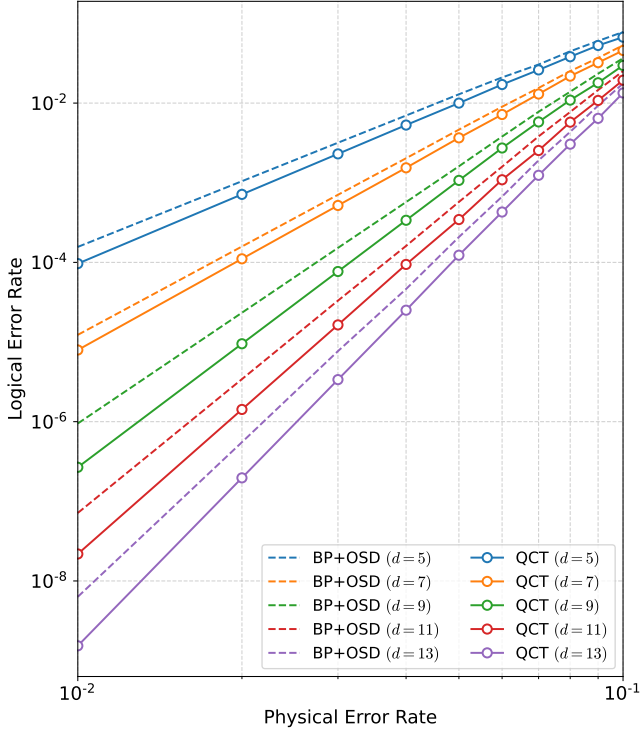


FIG. 8: Performance comparison of the proposed QCT decoder against the BP+OSD baseline.

Fig. 7 illustrates the decoding performance on surface codes with distances $d \in \{5, 7, 9\}$. Note that the total number of possible syndrome patterns the decoder should handle scales exponentially as 2^{d^2-1} . Despite this increasing complexity, QCT consistently outperforms the baseline decoders across all tested distances. Notably, the performance gap between QCT and the other methods widens as the code distance increases, highlighting the scalability of the proposed structure-aware architecture. These results demonstrate that, just as transformer architectures have shown superiority over FFNNs and CNNs in various domains, they also exhibit strong potential and serve as a promising architecture for QEC.

To further validate its performance, we compare QCT with the BP+OSD algorithm, a widely adopted baseline [20, 31–33]. For this comparison, we employ the quaternary BP [21] with maximum 20 iterations. As demonstrated in Fig. 8, QCT outperforms BP+OSD by a substantial margin across all code distances. These results confirm that QCT achieves superior decoding performance, outperforming both advanced neural decoders and the BP+OSD baseline. By applying qubit-centric embedding rather than stabilizer-centric perspective, QCT effectively captures the complex topological correlations of errors, establishing a new benchmark for quantum error correction decoding.

B. Analysis of Model Architecture

In this subsection, we investigate the impact of various architectural components—the number of transformer blocks, the structure-aware mask, and the merging layer—on the decoding performance of QCT for $d = 11$ surface code.

Number of Transformer Blocks: We examine the decoding performance according to the number of transformer blocks N . As shown in Fig. 9(a), increasing the model depth from $N = 2$ to $N = 8$ yields significant improvements in the LER; however, the performance gains saturate beyond $N = 8$. The substantial performance gap according to N demonstrates that a shallow architecture with only $N = 2$ blocks is insufficient to estimate the logical error present in the syndrome. Based on this observation, we select $N = 8$ as the default configuration, as it provides a balanced trade-off between performance and efficiency.

Structure-aware Masking: To validate the effectiveness of the structure-aware mask, we conduct an ablation study comparing the proposed QCT with an unmasked variant where all tokens globally attend to each other. Fig. 9(b) compares QCT models with and without structure-aware masking. The model without masking (labeled “w/o mask”) successfully learns the decoding task, indicating that the self-attention mechanism can implicitly capture error correlations. However, the models with structure-aware masking consistently outperform the model without masking across all tested code distances. This result highlights that while the standard transformer successfully learns the error patterns via global attention, explicitly concentrating on physically relevant neighbors is a significantly more effective strategy. The mask enables the model to focus on the topological structure of the codes, thereby enhancing the decoding performance.

Merging Layer: To validate the effectiveness of the merging layer, we compare QCT with the model without the merging layer (labeled “w/o merg”). The only difference is the existence of the FC layer that projects the concatenated $2D_e$ -dimensional vectors into D_m -dimensional unified tokens. Fig. 9(c) illustrates the decoding performance of the models with and without the merging layer.

As shown in the results, an additional merging layer provides a consistent improvement in decoding performance. Considering that this layer introduces a small number of parameters relative to the overall QCT architecture, achieving such a reduction in the LER is efficient. In quantum error correction, even these marginal gains are crucial for approaching the fault-tolerance threshold. Furthermore, it is worth noting that even the model without the merging layer still maintains excellent performance and outperforms the other baseline decoders. It demonstrates that shifting the decoding perspective directly to the qubit-centric approach is fundamentally effective.

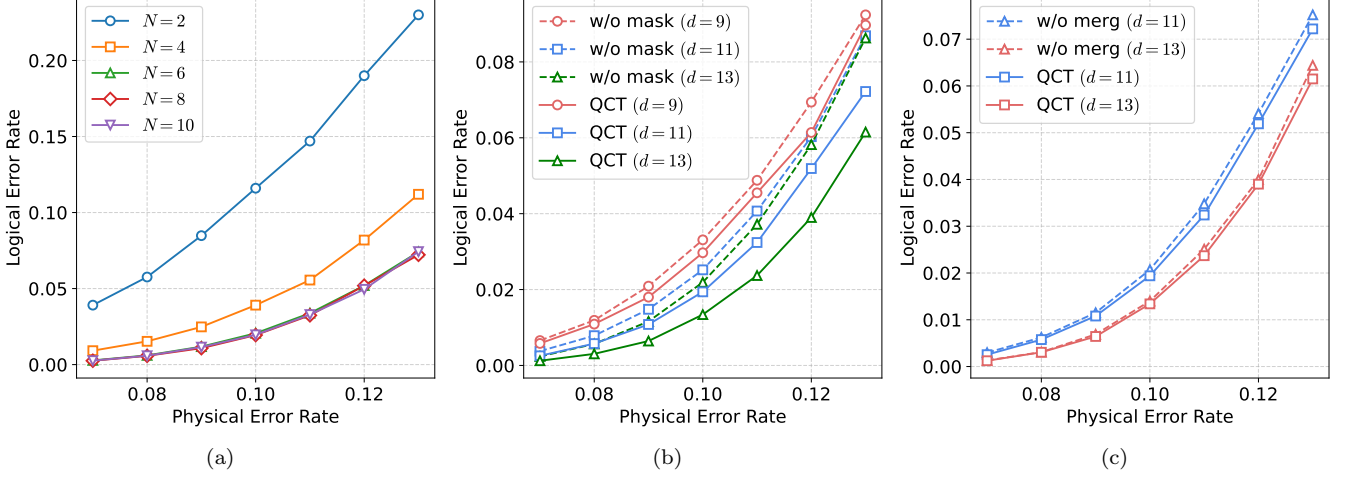


FIG. 9: Decoding performance of QCT under various configurations: (a) impact of the number of transformer blocks (N), (b) effect of structure-aware masking, and (c) effect of the merging layer to decoding accuracy.

TABLE I: Comparison of the pseudo-thresholds for the proposed QCT against benchmark decoders (MWPM, FFNN, and CNN) on the surface code for distances $d = 3, 5$, and 7 under the depolarizing noise model. The results of MWPM, FFNN, CNN are from [18].

Code distance	$d = 3$	$d = 5$	$d = 7$
MWPM [30]	8.28%	10.36%	11.94%
FFNN [13]	9.77%	11.35%	12.49%
CNN [18]	9.80%	12.15%	13.26%
QCT (Proposed)	9.80%	13.27%	14.31%

C. Threshold Analysis

In this subsection, we evaluate the pseudo-threshold and the threshold performance of QCT.

Pseudo-threshold: To further quantify the decoder's performance, we compute the pseudo-threshold, defined as the physical error rate at which the LER equals that of un-coded qubits [34]. As presented in Table I, for a small distance code ($d = 3$), the pseudo-threshold of QCT is comparable to that of other neural network decoders and substantially higher than that of MWPM. However, the performance gap widens with the code distance; for $d = 5$ and $d = 7$, QCT achieves the highest pseudo-threshold. This result demonstrates that QCT not only offers stronger error-correction capability but also scales more effectively with increasing code size.

Threshold: The code threshold is determined by identifying the crossing point of the LER curves for code distances ranging from $d = 5$ to $d = 13$. Fig. 10 compares the LER performance and threshold of QCT with BP+OSD decoder. As shown in Fig. 10(a), QCT achieves a high threshold of 18.1% under the depolariz-

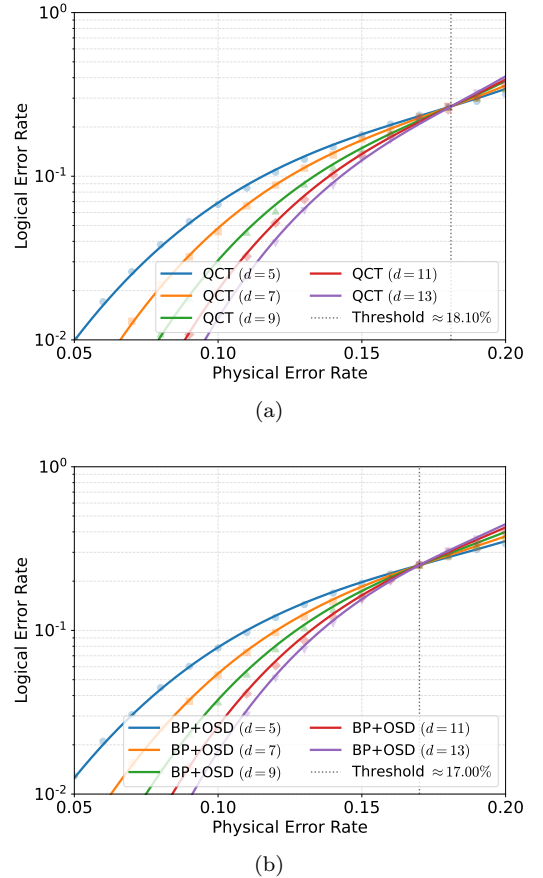


FIG. 10: Threshold performance for surface code with $d = 5, 7, 9, 11, 13$. (a) QCT. (b) BP+OSD.

ing noise model, which closely approaches the theoretical bound of 18.9% [27]. This value outperforms both the BP+OSD decoder with a threshold of 17.00% as shown

in Fig. 10(b), and the conventional MWPM decoder [26] with a threshold of 14.7%. The superior threshold of QCT indicates that the qubit-centric representation and structure-aware attention mechanism effectively capture complex error correlations.

V. CONCLUSION

In this paper, we proposed QCT, a novel decoder architecture that shifts the decoding paradigm from a stabilizer-centric to a physical qubit-centric perspective. Unlike syndrome-centric approaches in transformer-based QEC decoders, our method employs a qubit-centric representation via a merging layer. Furthermore, QCT fully utilized the topological structure of surface codes

by applying structure-aware masking, which effectively captures complex error correlations. Extensive simulations demonstrate that QCT consistently achieves superior decoding performance compared to existing neural network-based decoders and classical algorithms. Notably, QCT achieves an impressive threshold of 18.1% under depolarizing noise, significantly surpassing the thresholds of BP+OSD (17.00%) and MWPM (14.7%). Our analysis confirms that the synergy between the direct qubit-level representation and structure-aware masking is central to the model's superior performance. This work demonstrates that a transformer architecture grounded in a physical qubit-centric framework is highly effective for QEC decoding, paving the way toward practical, high-performance neural decoders for future fault-tolerant quantum computing.

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